



HOME / ABOUT

○ ABOUT HERDER ELEKTRONISCHE SYSTEMEN

Engineering systems that can be understood, *tested and trusted.*

Herder Elektronische Systemen is an independent engineering practice focused on embedded controls, power electronics, real-time testing, technical recovery and AI-assisted engineering software.

01 / ORIGINS

A practice built from *engineering work.*

Herder Elektronische Systemen grew from a working engineering test facility used for power electronics, embedded control, electric machines, instrumentation and real-time testing.

Development boards, power stages, motors, sensors, test equipment and experimental circuits became an environment for building complete systems. Hardware was

connected to firmware, firmware to controls, controls to instrumentation and measurements back to design decisions.

That engineering test work established the practice's central method: follow the technical problem across the full system until the implementation, physical behavior and evidence agree.

As projects became larger, another problem became equally important. Engineering context was often scattered across source files, models, test data, schematics, archived revisions and incomplete documentation. Recovering the reasoning behind a project could become as difficult as building it.

That need led to work in technical recovery, evidence preservation and AI-assisted engineering software. SafeFileAI and SafeECADAI emerged from the need to keep engineering conclusions connected to the files, measurements and decisions that support them.

THE ENGINEERING FOUNDATION

- Power-electronics prototyping
- Embedded control hardware
- Electric-machine test systems
- PCB development and bring-up
- Oscilloscopes, electronic loads and instrumentation
- Real-time control and automated data collection
- Hardware, firmware and model integration

Engineering across *connected systems.*

Herder Elektronische Systemen works across embedded software, electronics, controls, mechanics and test because the behavior of a prototype rarely belongs to only one layer.

A control issue may originate in sensing, timing, power delivery or mechanical response. A firmware problem may appear only under real hardware conditions. A test result may depend as much on instrumentation and setup as on the design being evaluated.

The practice therefore treats hardware, firmware, models, communications and measurements as parts of one engineering system. The objective is to identify the relevant interfaces, establish what the evidence supports and leave a technical path that can be reviewed and continued.

03 / TECHNICAL FOUNDATION

A capability *profile.*

EMBEDDED

Embedded C and C++
Firmware architecture
State machines
Timing and interrupts

CONTROLS

Motor control
Field-oriented control
Real-time systems
Feedback integration

COMMUNICATIONS

CAN
UART
SPI
I²C

MODELING AND ANALYSIS

MATLAB
Simulink
Python processing
Structured test data

HARDWARE

PCB design
Prototype bring-up
Sensors and actuators
Power-stage integration

POWER ELECTRONICS

Converter prototyping
Flyback development
Custom magnetics
Bench measurement

TESTING

Instrument integration
Automated data collection
Validation workflows
Technical reporting

RECOVERY

Repository assessment
Build reproduction
Evidence recovery
Technical documentation

SOFTWARE

AI-assisted engineering tooling
Evidence-aware technical workflows

04 / ENGINEERING PRINCIPLES

Evidence before *claims.*

01

Base claims on evidence.

02

Preserve system context.

03

Make changes traceable.

04

Test the integrated system.

05

State uncertainty clearly.

06

Leave the work easier to inspect and continue.

GOOD-FIT ENGAGEMENTS

Technical work with *enough substance.*

The strongest fit includes:

- Embedded or electromechanical prototypes
- Incomplete or inherited engineering projects
- Firmware and hardware integration
- Test automation and instrumentation
- Power-electronics prototypes
- Design recovery and technical documentation
- Technically demanding early-stage product development

05 / FOUNDER AND ENGINEERING LEAD

Jonathan Folkema

Jonathan Folkema is a mechanical engineer whose work spans embedded control, power electronics, electric machines, real-time testing, instrumentation and engineering software.

His project experience includes permanent-magnet motor drives, field-oriented control, CAN-connected embedded platforms, precision-motion equipment, PCB development, converter prototyping, automated data collection and recovery of complex technical repositories.

He founded Herder Elektronische Systemen to work on the interfaces that often determine whether a prototype can be understood, tested and continued.

A control problem may require inspecting the power stage. A firmware problem may originate in sensing or timing. A weak test result may come from instrumentation rather than the design itself. A project may appear complete but remain difficult to continue because its files, assumptions and evidence are disconnected.

The practice is built around hands-on investigation, measured validation, careful recovery and documentation that preserves what is known, what remains uncertain and what should happen next.

○ START A TECHNICAL DISCUSSION

Start with the *technical question.*

Describe the system, its current stage, the problem you are trying to resolve and what evidence is available.

Secure file exchange can follow the initial discussion.

START A TECHNICAL DISCUSSION →

Or write directly: info@herdersystemen.com



HERDER
ELEKTRONISCHE SYSTEMEN

© 2026 Herder Elektronische Systemen

[Privacy](#) [Engagement notes](#) [Back to top ↑](#)